

Discover

Trusted Products to eXpedite Your Discovery & Development – Advance Confidently



- Reflective of MOA of the drug molecule
- Specificity stability-indicating
- Relative potency range (50%-150%)
- Repeatable, robust, precise, and accurate
- Reproducible
 - Intra-lot (vial-to-vial)
 - Inter-lot (lot-to-lot)
 - Analyst-to-analyst
- Long-term support
 - Dedicated well characterized cell banks
 - Supply agreements for risk mitigation
 - Traceability and quality (audit-ready)



ICH-based Assay Qualification



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Assay Qualification of cAMP Hunter™ Tirzepatide GIPR Bioassay

Qualified with Tirzepatide

Optimized Dose Curve and Plate Layout for Qualification of the cAMP Hunter™ Tirzepatide (GIP RA) Bioassay

Dilution Scheme

For Tirzepatide	
Dilution Scheme	Concentration, ng/mL
	50
3	16.6667
3	5.5556
3	1.8519
2	0.9259
2	0.4630
2	0.2315
2	0.1157
2	0.0579
4	0.0145
4	0.0036

Representative Dose Curve (100% NC)

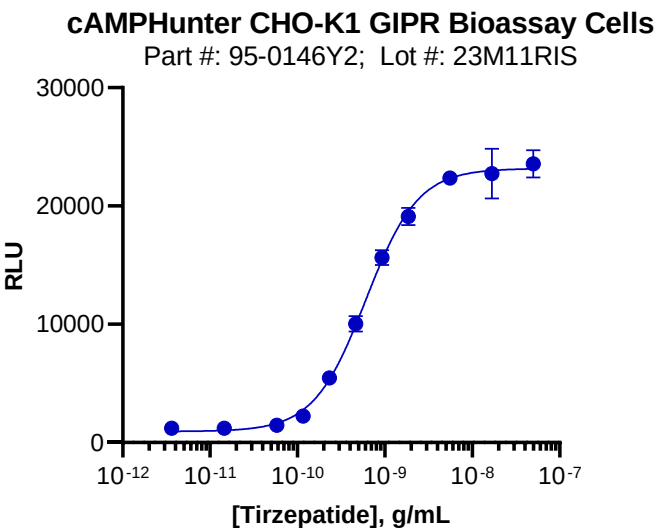
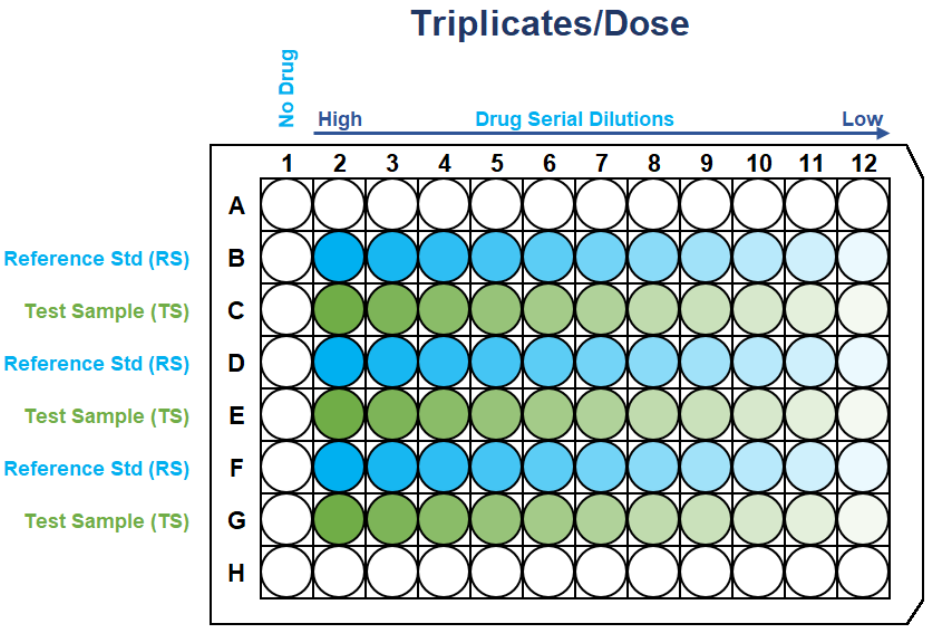


Plate Layout



RA = receptor agonist. NC = nominal concentrations

Day	Nominal Concentrations	Analyst
1	100% x 6 (repeatability)	1
2	200%, 140%, 70%, 50%	1
2/3	200%, 100%, 50%	2
3	200%, 140%, 70%, 50%	1
4	140%, 70%	2
4	200%, 140%, 70%, 50%	1
5	200%, 140%, 70%, 50%	1
6	200%, 140%, 70%, 50%	2

Evaluated 5 nominal concentrations (NC) over a range of 50%-200%

- (n=6 for each NC)

Repeatability: 6 runs of 100% NC by single analyst

Intermediate precision incorporates:

- 2 analysts
- Multiple days
- 4 NC
- 2 lots of bioassay cells

100% success rate (e.g. no plates failed system and sample suitability criteria)

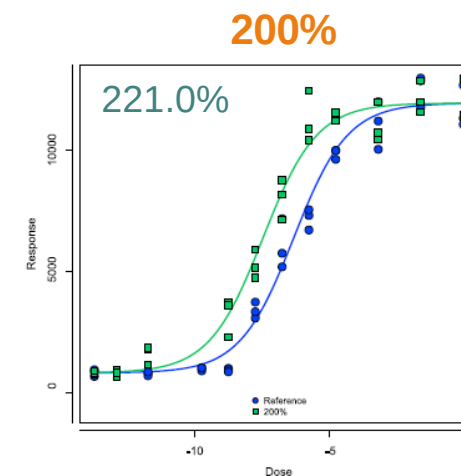
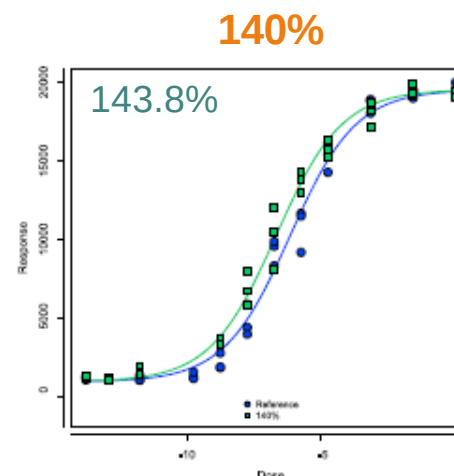
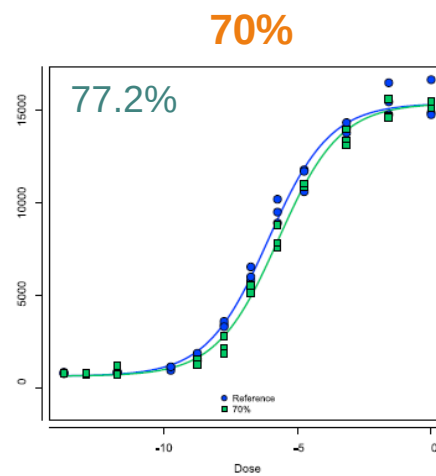
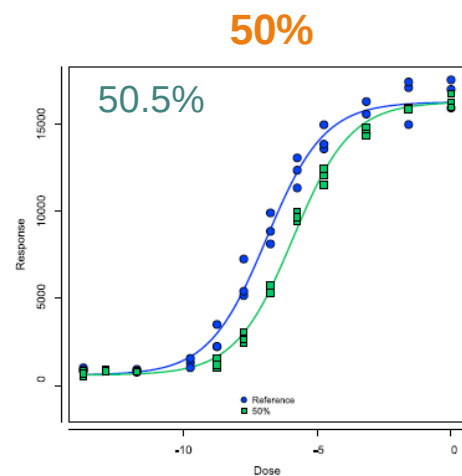
Assessment of Repeatability with 100% NC (Single Analyst)

Expected RP, %	Analyst	Measured RP, %	Potency Range, 95% CI, %	Mean RP, %	% RSD
100	1	106.621	0.93487 - 1.21599	103.0	12.4%
	1	111.123	0.93401 - 1.32208		
	1	96.097	0.82442 - 1.12015		
	1	119.955	1.04394 - 1.37835		
	1	101.467	0.88839 - 1.15890		
	1	82.994	0.73412 - 0.93828		

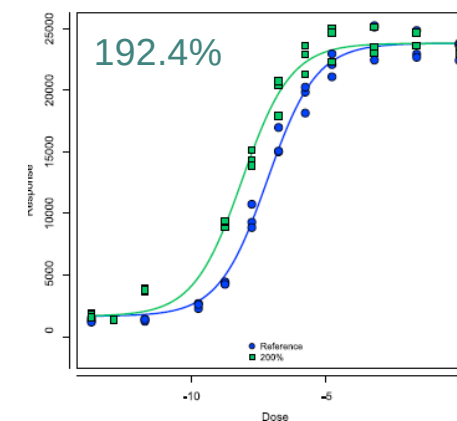
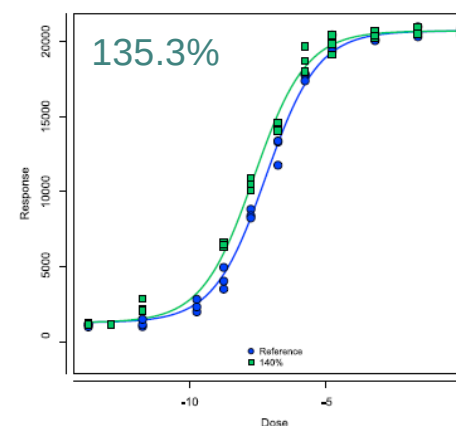
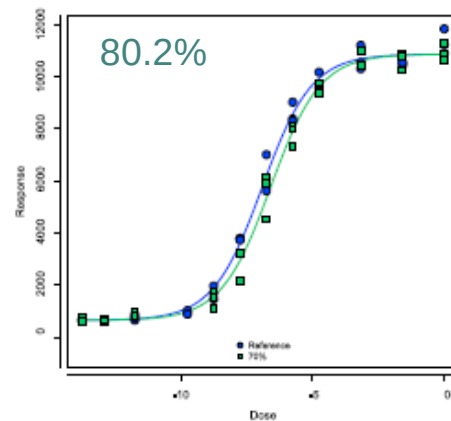
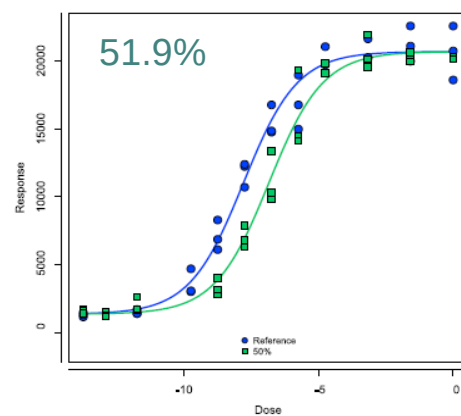
Representative Relative Potency Data

Analyst 1 vs Analyst 2: Four NCs Over a Range from 50% to 200%

Analyst 1

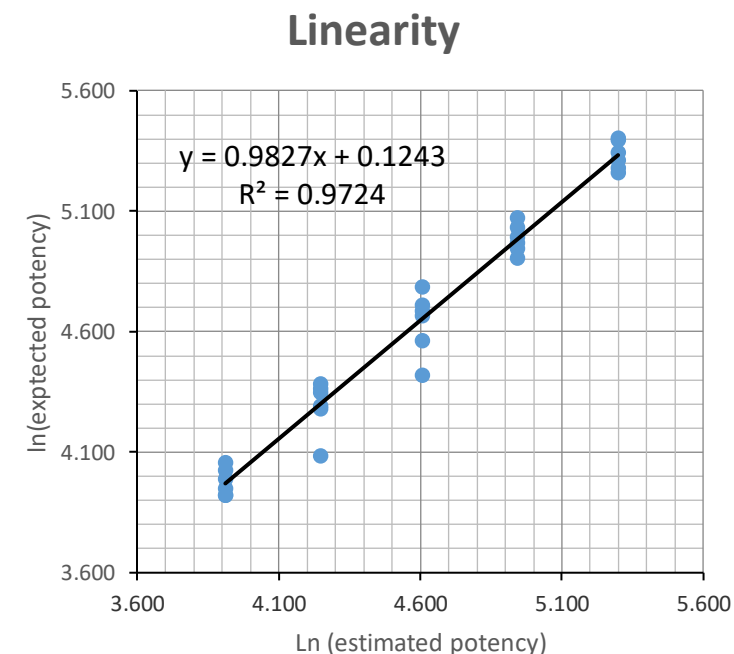


Analyst 2



Summary of Bioassay Qualification Data with Tirzepatide: Accuracy, Intermediate Precision and Dilutional Linearity

Expected Potency [%]	Exp.	Estimated Potency [%]	Recovery [%]	Acceptance criterion	Mean Recovery [%]
50	1	57.746	115	pass	107
	2	50.477	101	pass	
	3	56.0601	112	pass	
	4	51.905	104	pass	
	5	50.477	101	pass	
	6	54.074	108	pass	
70	1	78.667	112	pass	105
	2	77.25	110	pass	
	3	80.211	115	pass	
	4	72.267	103	pass	
	5	59.438	85	pass	
	6	73.109	104	pass	
100	1	106.621	107	pass	104
	2	111.123	111	pass	
	3	82.994	83	pass	
	4	108.522	109	pass	
	5	96.097	96	pass	
	6	119.955	120	pass	
140	1	153.339	110	pass	105
	2	143.855	103	pass	
	3	135.268	97	pass	
	4	147.404	105	pass	
	5	159.712	114	pass	
	6	140.89	101	pass	
200	1	222.775	111	pass	104
	2	221.027	111	pass	
	3	192.432	96	pass	
	4	209.417	105	pass	
	5	196.346	98	pass	
	6	202.525	101	pass	



Parameter	Value	Specification
Accuracy (Average % Recovery)	105%	100% +/- 20%
Repeatability	12.4%	≤20%
Intermediate Precision	≤8.7%	≤20%
Linearity (R ²)	0.972	≥0.95



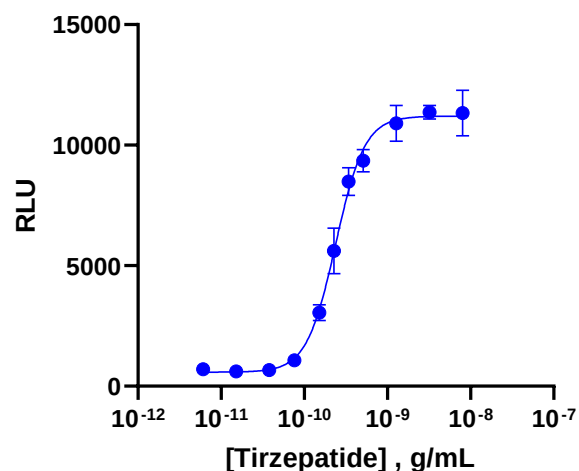
Assay Qualification of cAMP Hunter™ Tirzepatide GLP-1R Bioassay

Qualified with Tirzepatide

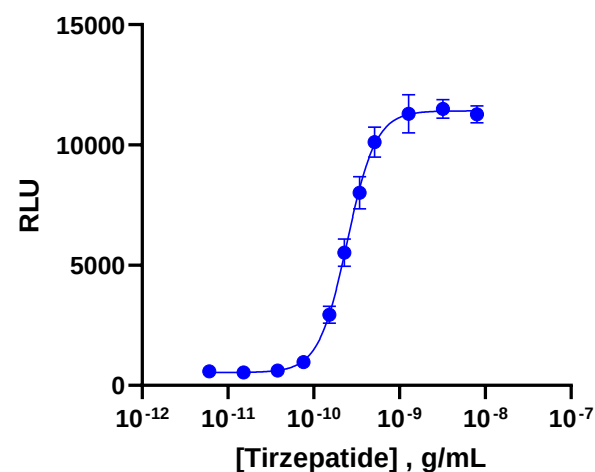
Excellent Inter-day Assay Reproducibility with Tirzepatide

Single Analyst, Multiple Days, Two Lots

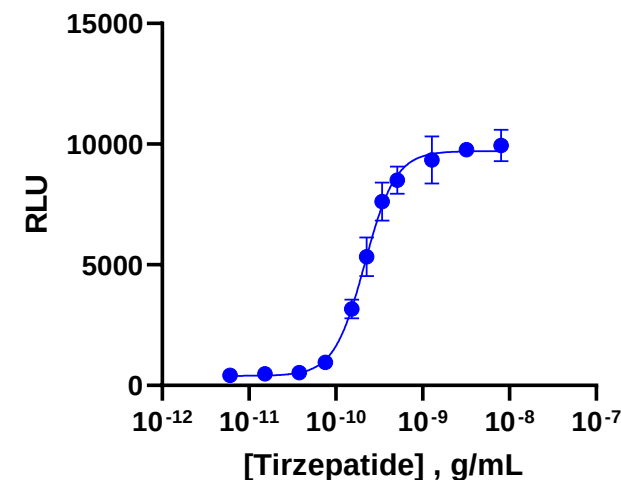
Day 1 (Lot-1)



Day 2 (Lot-1)



Day 3 (Lot-2)



	Day 1	Day 2	Day 3	% RSD
S/B	19.2	21.3	24.4	12.09%
EC ₅₀ , pg/mL	238.4	245.8	215.6	6.74%

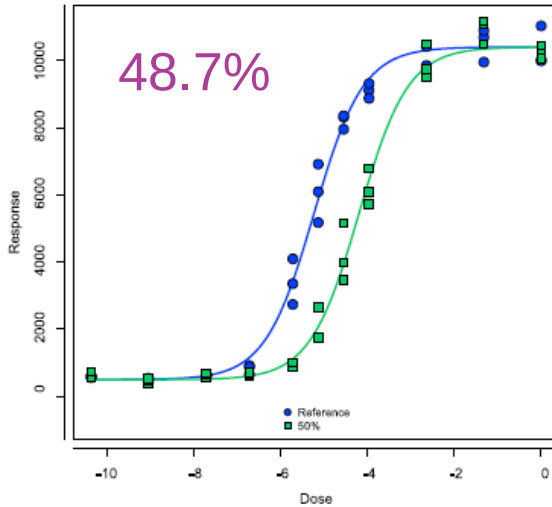
Consistent data generated with two different bioassay lots over multiple days

Representative Relative Potency Data (4 Nominal Concentrations)

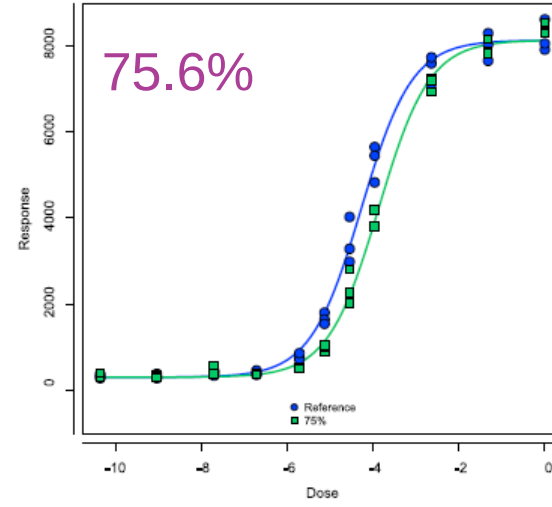
Analyst 1 vs Analyst 2

Analyst 1

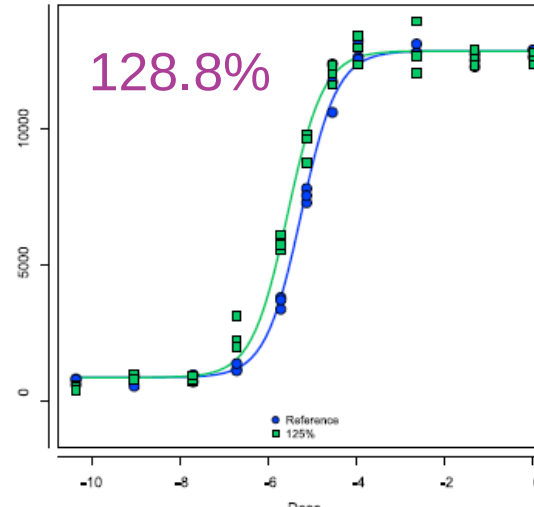
50%



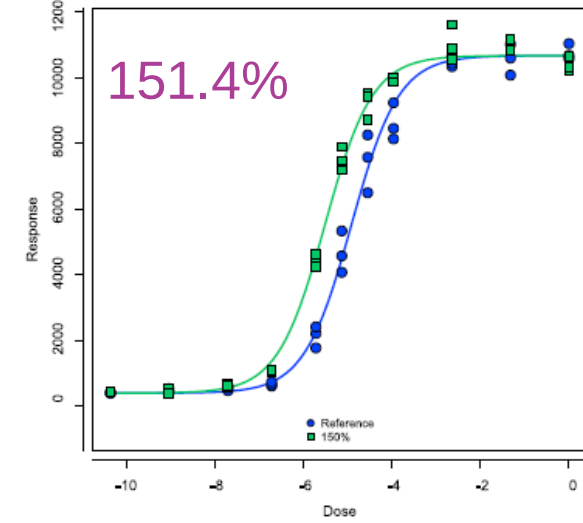
75%



125%

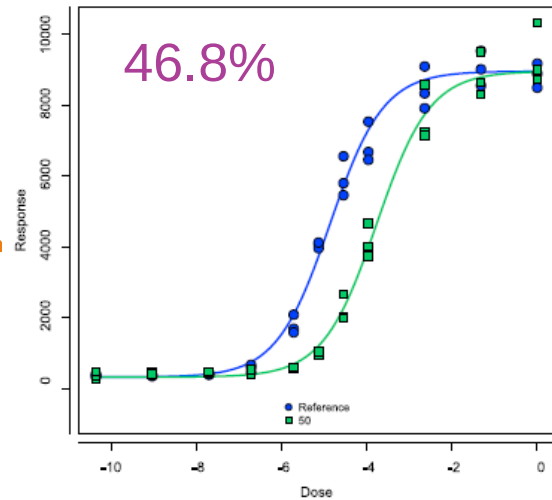


150%

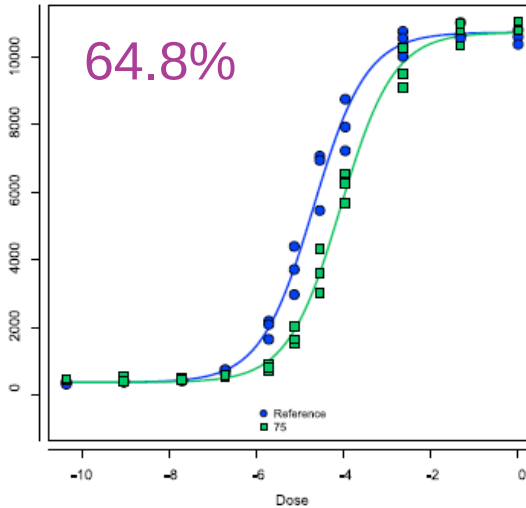


Analyst 2

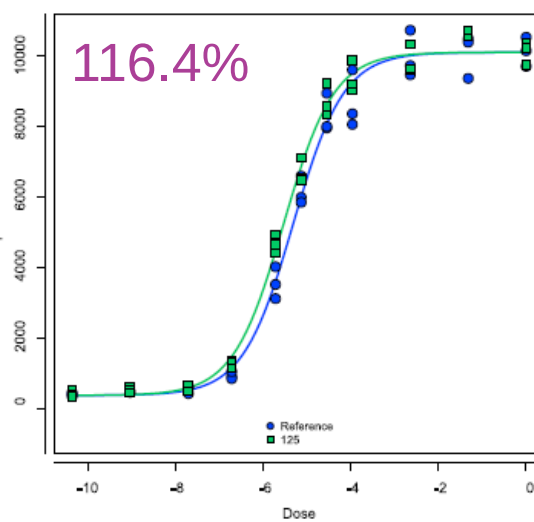
46.8%



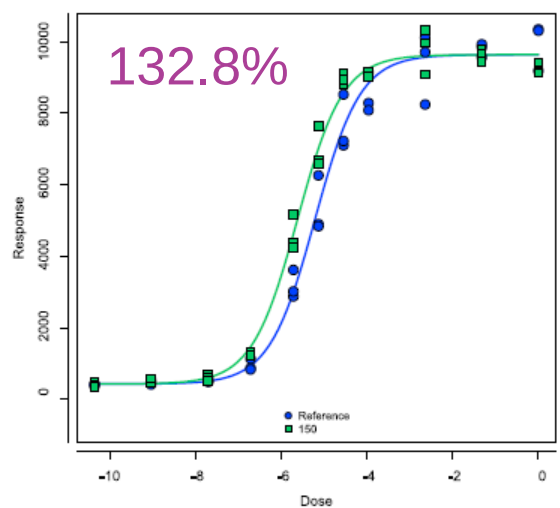
64.8%



116.4%

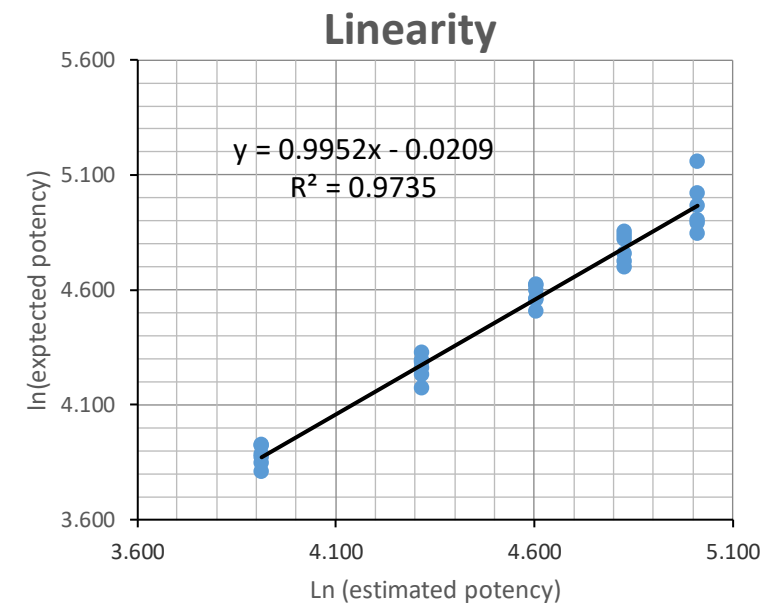


132.8%



Precision, Accuracy, and Dilution Linearity

Expected Potency [%]	Exp.	Estimated Potency [%]	Recovery [%]	Acceptance criterion	Mean Recovery [%]
50	1	50.394	101	pass	97
	2	48.735	97	pass	
	3	50.657	101	pass	
	4	47.943	96	pass	
	5	45.069	90	pass	
	6	46.819	94	pass	
75	1	75.63	101	pass	95
	2	70.862	95	pass	
	3	72.603	97	pass	
	4	73.186	98	pass	
	5	68.815	92	pass	
	6	64.854	87	pass	
100	1	101.832	102	pass	97
	2	95.68	96	pass	
	3	95.25	95	pass	
	4	99.167	99	pass	
	5	101.354	101	pass	
	6	90.489	90	pass	
125	1	123.808	99	pass	96
	2	112.588	90	pass	
	3	125.835	101	pass	
	4	128.026	102	pass	
	5	109.921	88	pass	
	6	116.357	93	pass	
150	1	174.066	116	pass	96
	2	151.405	101	pass	
	3	127.229	85	pass	
	4	143.271	96	pass	
	5	134.719	90	pass	
	6	132.764	89	pass	



Parameter	Value	Specification
Accuracy (Average % Recovery)	96.0%	100% +/- 20%
Repeatability	4.50%	≤20%
Intermediate Precision	≤6.6%	≤20%
Linearity (R ²)	0.973	≥0.95